Muhammad Subhan Siddiqui

subhansiddiquiau@gmail.com | 03066289689 | Islamabad, Pakistan

Kaggle | GitHub | LinkedIn

PROFESSIONAL SUMMARY

Result-driven Software Engineer specializing in **Full-Stack Development (MERN)** and **Applied AI/ML architecture**. Proven expertise in engineering end-to-end intelligent systems, including **fine-tuned Transformers**, deep learning-based **image restoration models**, and **production-ready cloud deployments**. Adept at optimizing model latency and accuracy, implementing robust backend infrastructures, and solving complex algorithmic challenges to deliver scalable, real-world engineering solutions.

EXPERIENCE

Software Engineer Fellow - Dev Weekends Fellowship (Remote) Jun 2025 – Nov 2025

- Solved 70+ LeetCode problems to strengthen Data Structures and Algorithms fundamentals.
- Engineered 4+ MERN stack projects in collaborative environments, gaining hands-on experience in scalable full-stack and backend development workflows.

PROJECTS

AI Financial Scam Detection Engine | GitHub | Deployed Link April 2026 – May 2026

- **Architected an advanced multi-layered hybrid AI system** combining a fine-tuned **DistilBERT** transformer for semantic intent classification, deterministic **spaCy** NLP pipelines for heuristic entity/claim extraction, and **Gemini 2.5 Flash** for contextual risk verification.
- **Engineered a custom weighted scoring algorithm** (0.70 * DistilBERT + 0.30 * Gemini) to calculate composite risk scores across continuous risk tiers, eliminating false positives on standard financial datasets (Financial PhraseBank).
- **Curated and preprocessed a high-variance master dataset** aggregating diverse NLP benchmarks (FakeNewsNet, FEVER, Clickbait) and integrated **synthetic data augmentation** to drastically reduce class imbalance and boost adversarial robustness.

PromptIR: All-in-One Image Restoration Model | GitHub May 2026 – Jun 2026

- **Fine-tuned PromptIR (NeurIPS 2023)** using **PyTorch** on a custom-built, multi-degradation dataset derived from **DIV2K**, creating an all-in-one vision model capable of concurrently restoring blurred, noisy, rainy, and dusty images.
- **Achieved substantial quantitative restoration gains** across targeted degradation tasks, including a **+4.89 dB PSNR gain** (SSIM +0.07) on noisy imagery and a **+4.77 dB PSNR gain** (SSIM +0.09) on motion blur over baseline initializations.
- **Designed a synthetic degradation pipeline** using **OpenCV** and **Scikit-image** to synthetically simulate multi-environmental corruptions (Mixed Degradation), validating model performance via pixel-level evaluation metrics (**PSNR/SSIM**).

TECHNICAL SKILLS

Programming Languages: Python, C/C++, JavaScript, SQL,
Frameworks & Libraries: ReactJS, Node.js, Express.js, PyTorch, OpenCV, NumPy, Scikit-image, Matplotlib, Tailwind CSS
Databases: MongoDB, PostgreSQL
Tools & Platforms: Git, GitHub, Docker, Google Generative AI (Gemini API), VS Code, Visual Studio, Google Colab, Hugging Face Spaces

EDUCATION

Air University Islamabad, Pakistan | Bachelor of Science in Computer Science 2023 - 2027 (Expected)
Relevant Coursework: Object-Oriented Programming, Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Artificial Intelligence, Software Engineering, Data Science, Digital Image Processing, Full Stack Web Development
Activities: Microsoft Learn Student Ambassadors (MLSA), Google Developer Groups (GDG) on Campus

CERTIFICATIONS

Python – Kaggle **April 2026**
Intro to Machine Learning – Kaggle | *Supervised Learning, Decision Trees, Regression, Scikit-Learn* **May 2026**
AI for Everyone – DeepLearning.AI (Coursera) | *AI Strategy, ML Workflows, Business Applications* **April 2025**
The Bits and Bytes of Computer Networking – Google (Coursera) | *TCP/IP, DNS, Routing, Network Security* **May 2025**